Smart Timetable: Predicting When Each Course Will Be Offered to Pre-book Students' Next-Semester Schedule

Executive summary · Machine Learning I Final Project · Universidad EAN · Team Semestria (B. Benavides, P. Bustos)

The decision. EAN undergraduates must wait for the official timetable before planning, while seats in popular time slots fill quickly. We propose a tool that learns each student's usual time slots and pre-books the pending courses most likely to be scheduled in those slots. The student decides; the Registrar's office confirms.

Data and method. We used real data: eight official EAN undergraduate timetables (2022-2 to 2026-1, 7,155 class-group records). In this supervised learning problem, for each course and each of 18 day-and-shift slots, the target variable is whether the course will be offered in that slot next term. Every feature uses only earlier terms, which prevents data leakage. We compared four model families against two baselines using time-ordered cross-validation with four folds, and used 2026-1 as a one-time test set. We chose logistic regression, which tied with the best model and is easiest to explain.

Key results (test term 2026-1). The model ranks slots better than the baseline of repeating last term's slot (average precision 0.83 vs. 0.73). Its precision was 90% (95% CI 86% to 94%). However, its recall was only 35%, because the decision threshold, set on earlier and less stable terms, proved too cautious. Calibration confirms this: the model underestimated probabilities in 2026-1. If a false positive (a wrong booking) costs twice as much as a false negative (a missed booking), the model's expected cost was higher than the baseline's (18.8 vs. 13.4 per 100 decisions). Both clearly beat doing nothing (25.9).

Recommendation. Run a pilot with the tool as an assistant, not an autopilot. It should book only high-confidence matches, suggest the baseline elsewhere, and send Saturday classes and newly coded courses to manual review.

Limits and risks. The model predicts whether a section will be offered, not whether a seat will be available. Its recall is lowest for evening and Saturday slots, likely used by working students, and it cannot predict courses with new codes. We will track precision and recall by shift every term, retrain and recalibrate the threshold each semester, and measure the real cost of wrong bookings.